

OVERVIEW

This document will explain how the raised letters Y, A and R near the beginning of the K3 ciphertext may be a clue leading to various transposition matrices. One of these matrices involves the CT of both K3 and K4, which then leads to several involving only K4. One of the most interesting of these K4 transposition matrices is presented below the detail image of the raised Y, A and R.



?ZKSUQGOJ
GKQKLMBPO
FUAXLKARK
EQHTISTND
QPFTF^YSNH
IOKBW^AEUJ
UWDSX^RGVF
WTINBIGDU
JOZCSASKT
CKZPWRZBV
LBLWOURT

K4:

?OBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
TWTQSQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDZXTJCDIGKUHUAUEKCAR

Why look at transpositions of K4, when we know that NYPVTT=BERLIN with a 1:1 correspondence (N=B, Y=E, etc.)?? Because it is still possible that the method behind K4 may be transposition/substitution/reverse-transposition, or something like the VIC cipher which includes one or more transposition steps.

YAR in K3 + K4

Below is the Candle AY/Antipodes matrix. For details on how this matrix was constructed, see the file “Candle AY.docx” in the folder “Files > Personal Folders > Tracy Cobbs”. Note the presence of YAR in the bottom right. Note, in particular, that the Y and A come from K3, while the R comes from K4. This evokes the fact that the raised Y and A in the first line of K3 CT are adjacent to each other, but not adjacent to the raised R.

```
?enDyahRohnLsrhEocpTeoiBidyShnaIachTnreYuldslls
qheEnctAycrEiftBrspAmhhEwenAtamAtegYeerLbteEfoa
gtdHardPneoHmgfMfeuHeecDmriPfeiMehnLsstTrtvDohw

LlnOhsnOsmrWxmnEtprNgatIhnrArpeSlnnElebLpiiAcaeWmtw
SfiOdueTuaeOtoaRmaeErtnRtibSeddNiaaHttmStewPierOagr
?OBKRUXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJQSSEKZZWA

ndiTeenRahcTeneUdreTnhaEoetFolsEdtiWenhAeioYteY
iewFebaEctdDhilCeihSiteGoeaOsddRydlOritRklmLehaa
TJKLUDIAWINFBNYPVTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR
```

The K3 plaintext can be read out of this matrix by a methodical route. The route starts with the SL in the middle of the left column and progresses to the left, wrapping around as needed. In each column the order of extraction is bottom bigram, top trigram, middle bigram. The first pass through the grid extracting K3 plaintext is presented here in bold caps. Taking the letter of K4 that lies below each K3 bigram as the K3 plaintext is extracted, we get this string:

```
?ZKSUQGOJGKQKLMBPOFUAXLKARKEQHTISTNDQPFTFYSNHIOKBWAEUJUWDSX
RGVFWTINBIGDUJOZCSASKTCKZPWRZBVLBLWOURT
```

This string turns out to be the skip-47 transposition of the K4 CT (with the leading ? included). What clue(s) might have led us to look at a skip-47 transposition, independent of the pathway through the Candle AY/Antipodes matrix? Recall that the CIA was officially created by the National Security Act of 1947. And recall that when Sanborn reused the Kryptos CT on the Antipodes sculpture, he cut 47 lines of CT into that copperplate.

SKIP-47 PADDED TO VARIOUS WIDTHS

Now let’s look at the skip-47 string when written with various widths. It has long been noted that K4 when written with width 21 has some notable features, so let’s start our examination of the skip-47 string with that width.

< 2 1 >
? Z K S U Q G O J G K Q K L M B P O F U A
X L K A R K E Q H T I S **T N D** Q P F T F Y
S N H I O K B W A E U J **U W D** S X R G V F
W T I N B I G D U J O Z **C S A** S K T C K Z
P W R Z B V L B L W O U R T

The CUT and ADD that have been noted when writing K4 on a width of 7 appear in this matrix.

A width of 25 produces a matrix from which the original order of K4 CT can be read out via a route highly reminiscent of the 86x4 K3 matrix presented in the recently released FOIA doc “CIAKryptosSculpturePresentation.pdf” on pp. 36 and 90. (This 86x4 matrix – split into two halves – is presented below.) Oddly, and perhaps only coincidentally, the letters in the upper and lower right corners of both the width 21 and width 25 matrices are A and Z. The letters in the lower left are P and C, which (again, perhaps coincidentally) are the letters shown in the margin of Sanborn’s 42x8 K3 transposition worksheet.

< 2 5 >
? Z K S U Q G O J G K Q K L M B P O F U A X L K A
R K E Q H T I S T N D Q P F T F Y S N H I O K B W
A E U J U W D S X R G V F W T I N B I G D U J O Z
C S A S K T C K Z P W R Z B V L B L W O U R T
9 8 7 6 5 4 3 2 1

SLOWLYDESPARATLYSLOWLYTHEREMAINSOFPASSAGEDE . . .
ASREMOVEDWITHTREMBLINGHANDSIMADEATINYBREACH . . .
OLEALITTLEIINSERTEDTHECANDLEANDPEEREDINTHEH . . .
FLICKERBUTPRESENTLYDETAILSOFTHEROOMWITHINEM . . .
9 8 7

. . . BRISTHATENCUMBEREDTHELOWERPARTOFTHEDOORWAYW
. . . INTHEUPPERLEFTHANDCORNERANDTHENWIDENINGTHEH
. . . OTAIRESCAPINGFROMTHECHAMBERCAUSEDTHEFLAMETO
. . . ERGEDFROMTHEMISTXCANYOUSEEANYTHINGQ
6 5 4 3 2 1

YAR in K4

Skip-47 padded to a width of 9 produces the matrix presented on page 1, in which YAR appears in one of the columns. The positions of the 11 letters which Sanborn has revealed decode to BERLINCLOCK – NYPVTTMZFPK – have been highlighted in boldface.

```
< 9 >
? Z K S U Q G O J
G K Q K L M B P O
F U A X L K A R K
E Q H T I S T N D
Q P F T F Y S N H
I O K B W A E U J
U W D S X R G V F
W T I N B I G D U
J O Z C S A S K T
C K Z P W R Z B V
L B L W O U R T
```

Reading each column from top to bottom, taking each column in turn from left to right, yields the skip-31 transposition of K4.

This observation provides a convincing reason why the sculpture tableau, the final four lines of CT, and Jim's Sanborn's released worksheets all have 31 columns!

Other formattings of K4 skip transposition strings

Skip-31 on a width of 19:

```
? G F E Q I U W J C L Z K U Q P O W T
O K B K Q A H F K D I Z Z L S K X T T
B S N C P W U L L I F W X B S W O Q M
K S Y A R I A R U G B A T S E G G S Z
R O P R N N U V D K B T J O K D H J F
U T V
```

Skip-65 >

Skip-31 V alternate columns (1,3,5,2,4)

5 x 20

?**Y**K**R****N**

G**A**BNC

F**R**KNP

EIQ**U****W**

QAAVU

IRHDL

UUFKL

WGKBI

JBDTF

CAIJ**W**

LTZOX

ZSZKB

KELDS

UGSH**W**

QG**K**JO

PSX**F**Q

O**Z**T**U****M**

WR**T**TK

TOB**V**S

O**P**S

Note how the pattern of the 5 W's found in K4, when read from bottom to top in this matrix, is similar to the typography of K3's dYAhR substring. L-R-R-L-R = unraised-raised-raised-unraised-raised.